



CHAPTER 3

Helium Rich Mass Transfer



All hands, this is the captain. Begin prepping all stations for first contact.

— *Captain Ed Mercer* —



SINCE THE FORCE BETWEEN TWO objects due to gravity scales like $F \sim \frac{1}{r_{\text{sep}}^2}$, another question naturally arises during the consideration of contact binaries. For sufficiently close binaries, one can imagine that mass transfer through RLOF or tidal effects could become reasonably strong, stripping the hydrogen envelope of a main sequence (or shortly post-main sequence) star and exposing the helium-enriched core underneath. If this stripping does not cause the star to retreat within its Roche lobe, then this core material would continue to be accreted, creating helium-enriched matter streams accreting onto its companion.

As discussed in Section 2.4, the presence of a companion measurably alters the evolution of a star. Typically this manifests as allowing matter to be exchanged between the components or the gravity of one star warping the photosphere of the other. In this chapter we focus on the first instance, where mass transfer occurs between two stars. In particular, we consider a binary system with ZAMS masses of $30 + 10 M_{\odot}$, and an initial period of $10^{0.6} \approx 3.98$ d, and standard solar composition ($X = 0.7, Y = 0.28, Z = Z_{\odot} = 0.020$, with an equivalent abundance mix of elements as described in section 2.1.1 of [Eldridge et al. \(2017\)](#) which is in turn based on [Grevesse & Noels \(1993\)](#)). We specifically explore this case as a case study on the effect of helium enriched mass transfer. Future work will be needed to examine the prevalence of the effect of this mass transfer on a grid of close binaries.

3.1 | Motivation

The recent discovery of intermediate-mass helium stars in the “gap” between subdwarf helium stars and Wolf-Rayet (WR) stars ([Drout et al., 2023](#); [Götberg et al., 2023](#)) is a significant shift in the understanding of the importance of interacting binary stars for stellar populations. These stars have $\sim 8\text{--}25 M_{\odot}$ progenitors that have had their hydrogen envelope stripped via binary interaction. Although many groups have investigated the importance of these objects for stellar populations, such as sources of ionising photons (e.g. [Stanway et al., 2016](#); [Götberg et al., 2017](#); [Xiao et al., 2018](#)), hydrogen-poor core-collapse supernovae (e.g. [De Donder & Vanbeveren, 1998](#); [Smith et al., 2011](#); [Eldridge et al., 2013](#); [Eldridge & Maund, 2016](#); [Laplace et al., 2021](#)) and gravitational wave transients and/or γ -ray bursts (e.g. [Eldridge et al., 2011](#); [Qin et al., 2018](#); [Bavera et al., 2020](#); [Fuller & Lu, 2022](#); [Stevance et al., 2023](#)), until direct confirmation of their existence, uncertainty remained about their validity. Importantly, now with the potential detection of these objects, we can begin to test and verify our models of these stars.

The evolution of a star in a binary is complicated in many ways due to the presence of its companion. Notably, the two stars may exchange mass, altering the orbit dependent on the mass of the donor and accretor stars. Much study has been devoted to the nature of binary mass transfer (among others, [Delgado & Thomas, 1981](#); [Soberman et al., 1997](#); [Davis et al., 2013](#); [Landri et al., 2024](#)). In some studies, such as [Dutta & Klencki \(2023\)](#), which concentrate on the evolution of the donor star, the accretor is assumed to be a point mass. However, studies where the companion is followed in detail show that the evolution of the accretor can be exotic compared to that of a star from single stellar evolutionary pathways, (e.g. [Wellstein et al., 2001](#); [Renzo](#)

& Götberg, 2021; Renzo et al., 2023; Menon et al., 2024; Schneider et al., 2024; Wagg et al., 2024).

Most previous work has focused on the structure of the donor as it becomes a helium star. Epochs of mass transfer alter the structure of the constituent stars in a binary (Law-Smith et al., 2020; Klencki et al., 2021; Renzo & Götberg, 2021), but the accretor has only been investigated superficially (e.g., Packet, 1981; Cantiello et al., 2007; de Mink et al., 2013; Renzo & Götberg, 2021). These studies indicate that the accretor requires further investigation. The accretors might, as a group, provide the most unusual and exotic stellar objects we observe in the Universe, such as chemically homogeneous stars (de Mink et al., 2013; Ghodla et al., 2023) and certain types of chemically peculiar stars (e.g., Hunter et al., 2008; Langer, 2012). Thus, the accretors in a binary merit further investigation.

There are open questions in the evolution of massive binary stars that a further study of accretors may provide insights into. These include, for example, the discrepancy between the spectroscopic mass and the evolutionary mass (see, for example, Herrero et al., 1992; Massey et al., 2009; Bouret et al., 2013; Massey et al., 2013; McEvoy et al., 2015; Markova et al., 2018; Ramachandran et al., 2018; Bestenlehner et al., 2020). Mass transfer alone seems insufficient to explain this, given that most codes use an implementation or variant of the so-called “Dutch” mass loss scheme of de Jager et al. (1988); Vink et al. (2001). Rather, this discrepancy has commonly been attributed to deficiencies in the physics of model atmospheres (Herrero et al., 1992) or, more recently, missing or uncertain physics in the stellar evolution models (Bestenlehner et al., 2020). It has been observed through multiple modelling efforts such as BONNSAI (work in Bestenlehner et al. 2020, BONNSAI in Schneider et al. 2014), the work of Maeder (1990) (Herrero et al., 1992), the Geneva tracks (work by Markova et al. 2018, Geneva code by Ekström et al. 2012) and the PoWR code (work by Ramachandran et al. 2018, PoWR code in Gräfener et al. 2002; Hamann & Gräfener 2003; Sander et al. 2015).

Previous studies of close binaries suggest that they experience Case A mass transfer, i.e., mass transfer during the main sequence, and produce Algol systems in which the less massive component fills its Roche lobe (e.g. Leung, 1989; Pols, 1994; Vanbeveren et al., 1998; Wellstein et al., 2001; de Mink et al., 2007; Sen et al., 2022). In tight binaries, this can lead to a contact phase, with both stars filling their Roche lobes. Contact binaries were studied as early as the 1970s (Robertson & Eggleton, 1977), though it is only in the last few decades that progress has moved from the lower mass W Ursae Majoris binaries to massive contact binaries – Menon et al. (2021a) was the first, providing a study of the LMC and SMC. Most work focuses on the population properties of these contact binaries, with few in-depth analyses of individual systems. As in Sen et al. (2022), if the mass transfer rate exceeds the slow Case A regime (where mass transfer is on the nuclear time-scale) and enters the fast Case A regime (where it is on the thermal time-scale), the hydrogen-rich envelope of the star may become depleted; thus, the helium core is exposed and the material being accreted may become helium rich. An alternative scenario is a main sequence merger event if the orbit hardens sufficiently; this generally requires filling of the second Lagrange point as well (Tylenda et al., 2011; Nandez et al., 2014).

The effect of helium-enriched mass transfer has previously been studied (e.g., Delgado & Thomas, 1981). Most authors, however, treat the problem of Case BB helium mass transfer (mass transfer after the star has

lost its hydrogen envelope) as having originated from a helium star on the helium main sequence, rather than as emerging from previously hydrogen-dominant mass transfer. In the latter case, it is specifically binaries formed as a pair from *birth* that experience this mass transfer, with helium-rich material originating from core hydrogen burning in the donor star (e.g., [Renzo & Götberg, 2021](#)).

Mass transfer of particularly helium-rich material would result in a binary in which thermohaline mixing becomes important to include. This is the process by which an inverted mean molecular weight profile is close to its stability limit due to the temperature gradient (e.g. [Ulrich, 1972](#); [Kippenhahn et al., 1980](#); [Charbonnel & Zahn, 2007](#); [Angelou et al., 2011](#); [Fraser et al., 2022](#)). Thermohaline mixing was quantitatively introduced by [Ulrich \(1972\)](#) by analogy to a saline solution such as ocean water ([Stern, 1960](#)). The inclusion of the thermohaline mixing mechanism, in this context, allows material with a greater mean molecular weight to settle towards the core of the star, which one expects to become dominant during the accretion of helium-rich material. It is commonly included in most binary evolution models (e.g., [Fragos et al., 2023](#); [Sen et al., 2023](#)). However, it is difficult to directly constrain the strength of thermohaline mixing by direct measurement (e.g., [Stancliffe & Glebbeek, 2008](#); [Angelou et al., 2011](#); [Angelou et al., 2012](#), and references therein), predominantly because the mixing happens subsurface, and is observed phenomenologically by reference to abundance anomalies.

In this Chapter, we investigate both the impact of helium-rich material on an accretor star, and the thermohaline mixing that this induces, with an example binary stellar evolution model. Our aim is to demonstrate that it is important to account for the varying composition of material transferred between the donor and accretor in a binary. We also aim to determine the effect of the helium-rich material on the accretor and investigate the impact of thermohaline mixing on the resultant composition profiles created in the accretor.

3.2 | Numerical Method

Our general method remains the same as outlined in Chapter 2, and we use the AOTEAROA STARS code outlined in Section 2.6.1. We outline here the key changes made for this work.

During accretion, the infalling material may have a composition different from that of the accretor’s surface layer. We allow for *variable composition accretion* (VCA) in the following way. If one star is filling its Roche lobe, we set the abundance of the accreted material to the surface abundance of this star. If both stars are filling their Roche lobes, we set the abundance of the accreted material to the *average* of the two surface abundances of both stars. The primary reason for this modification is that if both stars are filling their Roche lobes, not averaging their abundances would mean that we would be artificially altering the composition of the accretor’s surface.

To evaluate the importance of this VCA, we also create a model where we do not link the surface composition of the accretor to that of the donor, with this being our *non-variable composition accretion* (NVCA) model. In these models, we set the abundance of the accreted material to that of the surface of the accretor,

which is in line with the default behaviour of MESA (Paxton et al., 2015).

This variable accretion will result in an inverted mean molecular weight profile, resulting in thermohaline mixing. To model this, we use the prescription of Kippenhahn et al. (1980) and Stancliffe et al. (2007). Thermohaline mixing in the accretor is treated as a diffusive process, where the diffusion coefficient D_{th} is given by

$$D_{\text{th}} = \alpha_{\text{th}} \frac{16aT^3 H_P}{(\nabla_{\text{ad}} - \nabla) c_P \rho \kappa} \left| \frac{d\mu}{dr} \right| \frac{1}{\mu}, \quad (3.1)$$

where H_P is the pressure scale height, a is the radiation pressure constant, ∇_{ad} is the adiabatic temperature gradient, ∇ is the temperature gradient, μ is the mean molecular weight, c_P is the specific heat at constant pressure, κ is the opacity, and ρ is the density. We also introduce α_{th} , a thermohaline mixing efficiency parameter. $\alpha_{\text{th}} = 0$ represents no thermohaline mixing, and $\alpha_{\text{th}} = 1$ represents efficient thermohaline mixing. We note that this definition differs slightly from the definition in other works, most notably MESA (Paxton et al., 2013). Our efficiency parameter can be converted to theirs using the relationship

$$\alpha_{\text{th}} = \alpha_{\text{th}}^{\text{MESA}} \cdot \underbrace{\sum_k \frac{1}{16} \frac{g_k}{P_k} (\mu_{k+1} + \mu_k) \left(\frac{\varphi}{\delta} \nabla_{\mu} - \nabla_{\text{ad}} \right) \left| \frac{\partial r}{\partial \mu} \right|_k}_{\equiv \mathcal{C}}, \quad (3.2)$$

where g is the local gravity, $\varphi = \frac{\partial \ln \rho}{\partial \ln \mu}$, $\delta = \frac{\partial \ln \rho}{\partial \ln T}$, r is the radius, and subscript k refers to a quantity internal to the meshpoint k . As this is meshpoint-dependent, and thus structure-dependent, in general the meshpoint sum will vary throughout the evolution of the star. At the onset of thermohaline mixing, the meshpoint sum in the presented model evaluates to $\approx 1.62 \times 10^{-4}$. In general throughout the VCA model the constant is between 1×10^{-6} and 1, and this is shown in Figure 3.1.

We note that our mass loss prescription is given by Eqn(2.18), with the choice of $k = 3 \times 10^{-3}$. This scaling factor is a combination of the efficiency parameters 3×10^{-6} from Hurley et al. (2002, equation 59) and 1000 from Claeys et al. (2014, equation 11). The first choice was made by Hurley et al. (2002) as a free parameter by experiment to ensure numerical stability. In the second, Claeys et al. (2014) included the additional factor to ensure that thermal time-scale mass transfer occurs with a higher mass-transfer rate while also retaining numerical stability. We note that this is an incomplete implementation of the Claeys et al. (2014) prescription of RLOF, and we do not interpolate between subunity and superunity mass ratios. We have run a model with the full implementation, and find the only difference in evolution is towards the end of evolution of the donor, where the evolution becomes numerically unstable. There is also an inconsistency because in contact systems the value of k given in Claeys et al. (2014) depends on the mass ratio – which means that it is different for the donor and accretor, even if they are both main sequence stars. To remove this inconsistency, we leave the value of k constant as it provides numerical stability in our models. We note that this should be replaced in the future by a more physically motivated prescription such as that given by Kolb & Ritter (1990).

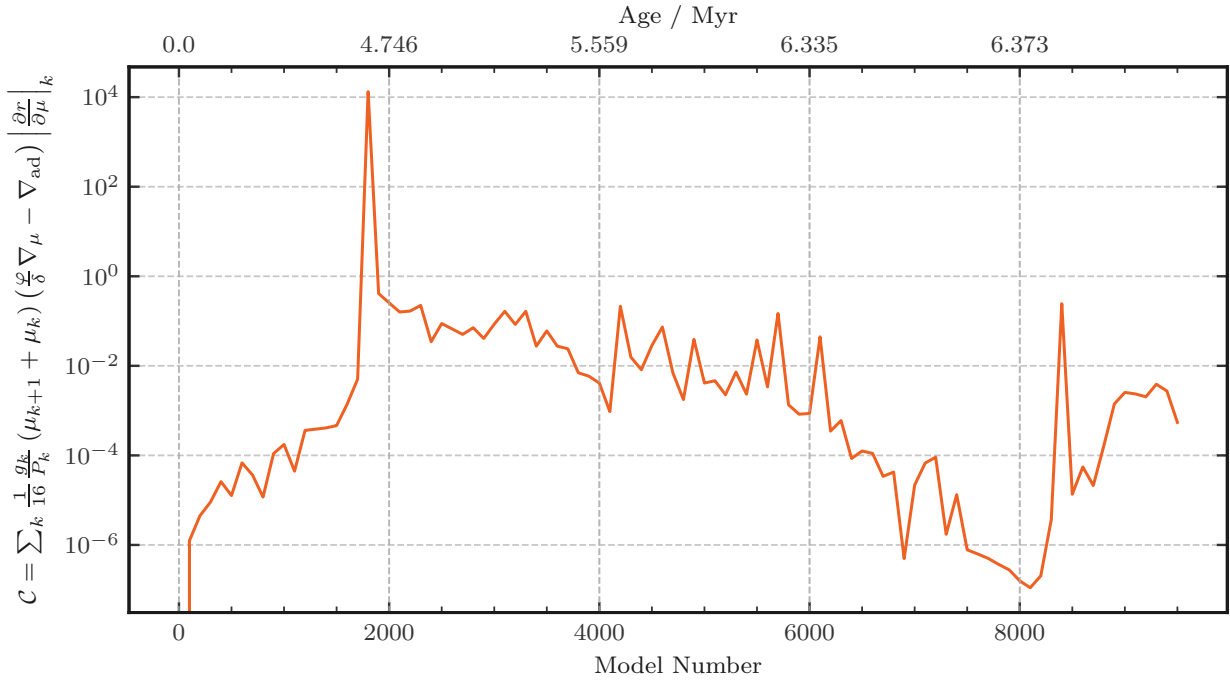


Figure 3.1: The value of the meshpoint sum $\mathcal{C}$ in Eqn(3.2) throughout VCA.

3.3 | Evolution overview

Stage	Letter	Section described
Model begins	A	Section 3.3.1
RLOF begins	B	Section 3.3.2
Contact begins	C	Section 3.3.3
Contact ends	D	Section 3.3.4
RLOF ends	E	Section 3.3.5

Table 3.1: The five major stages of the evolution of our binary.

The binary’s evolution can be broken up into five main stages, explored in the following section. Each stage is assigned a label, which is delineated in Table 3.1 – in brief, the lexicographic ordering reflects temporal evolution of the binary. That is to say, (A) represents the ZAMS, (B) the start of RLOF on the donor, (C) the start of RLOF for the accretor, (D) the end of RLOF for the accretor, and (E) the end of RLOF for the donor. Unless otherwise indicated, at every stage we shall be discussing the VCA case. The full evolution for the VCA (NVCA) accretion case is shown in Figure 3.2 (Figure 3.3), with subplots showing the behaviour around the contact phase.

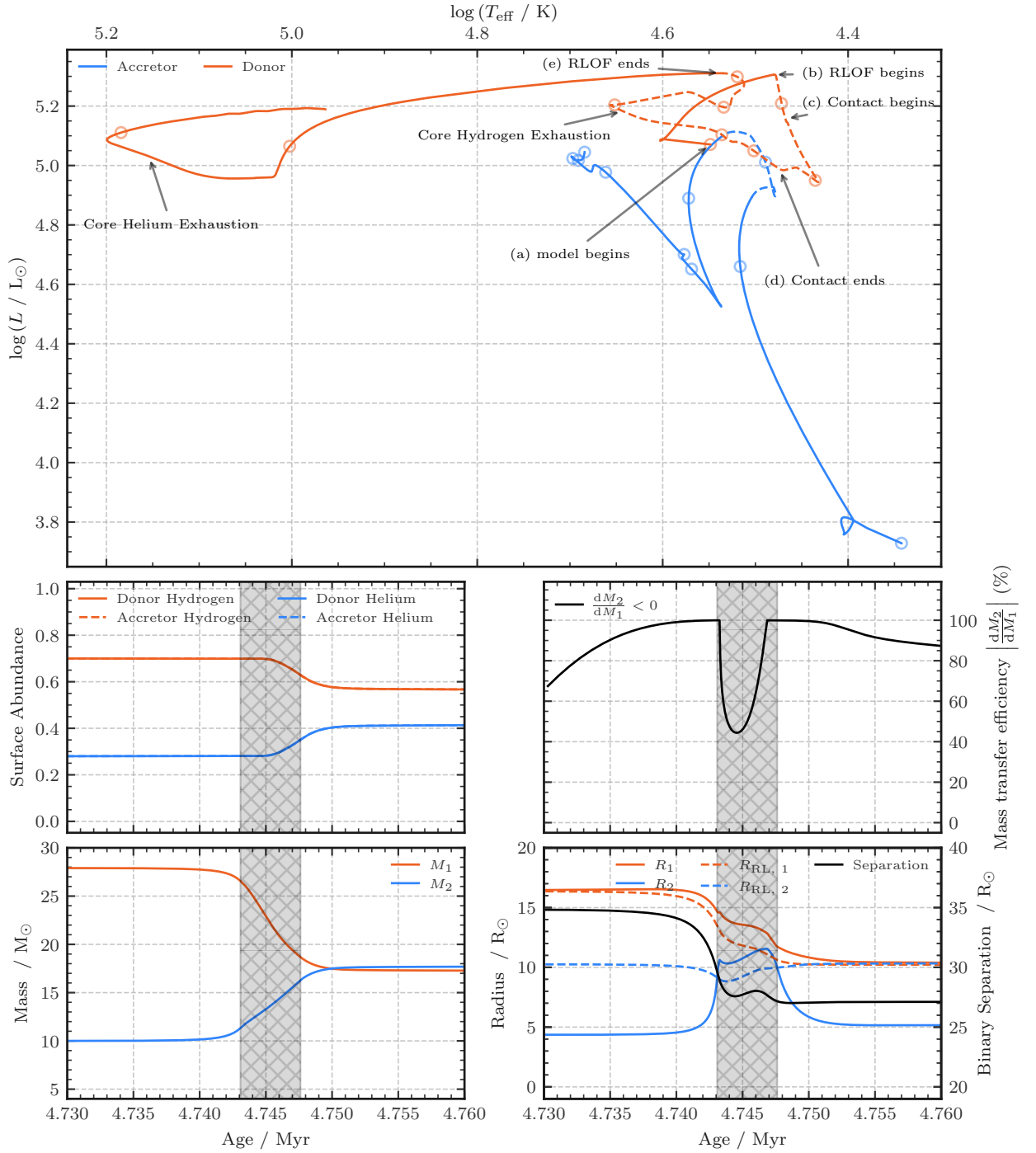


Figure 3.2: The main evolution of a $30 + 10 M_{\odot}$ binary with VCA enabled. Clockwise from top: the HRD with stages in Figure 3.7 and Table 3.2, the mass transfer efficiency, the separation / radius, the mass, and the surface abundances. In the lower four plots, the gray region shows the contact phase, and the plots are cropped to show the ~ 13 kyr on each side of the contact phase. The dashed line on each HRD track shows when the relevant star is overfilling $\mathcal{L}_1$, and contact is when both stars are dashed.

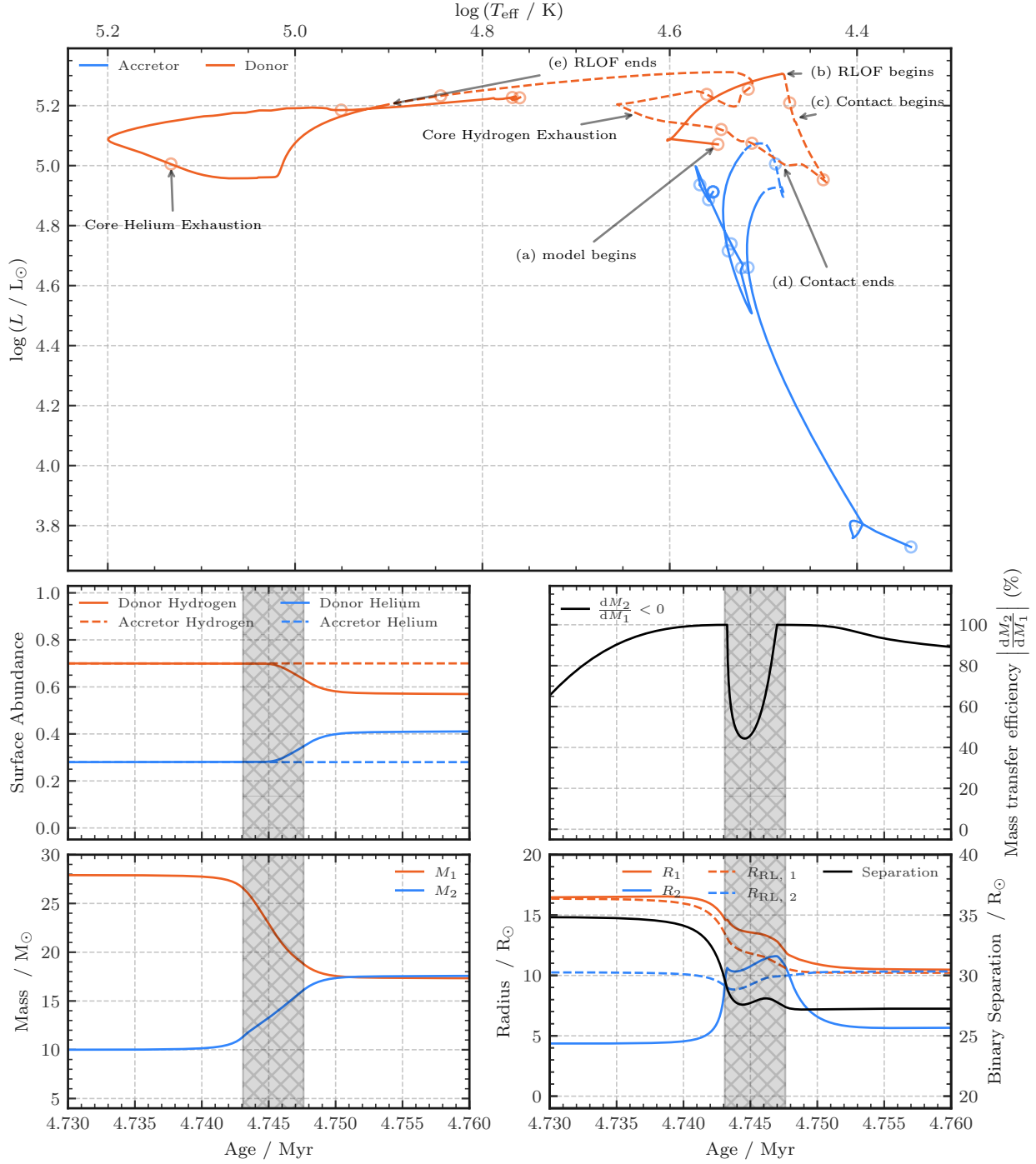


Figure 3.3: As for Figure 3.2, but for the NVCA model.

3.3.1 | Pre-interaction (stages A–B)

The pre-interaction phase of both stars is identical in both cases, as the key change enabling this work only begins when RLOF occurs. In both cases, it evolves as a fully detached binary for 4.51 Myr, with the donor almost completing its main sequence evolution.

3.3.2 | Pre-contact (stages B–C)

When the donor fills its Roche lobe, it begins to efficiently transfer material to its companion. The mass transfer shrinks the orbit by approximately 23 per cent. The accretor begins to accrete material, bleeding angular momentum into the system and causing the accretor to spin up. As the contact phase nears, the accretor approaches 40 per cent of the critical rotation rate, as shown in Figure 3.4. It reaches approximately 20 per cent when the contact phase begins.

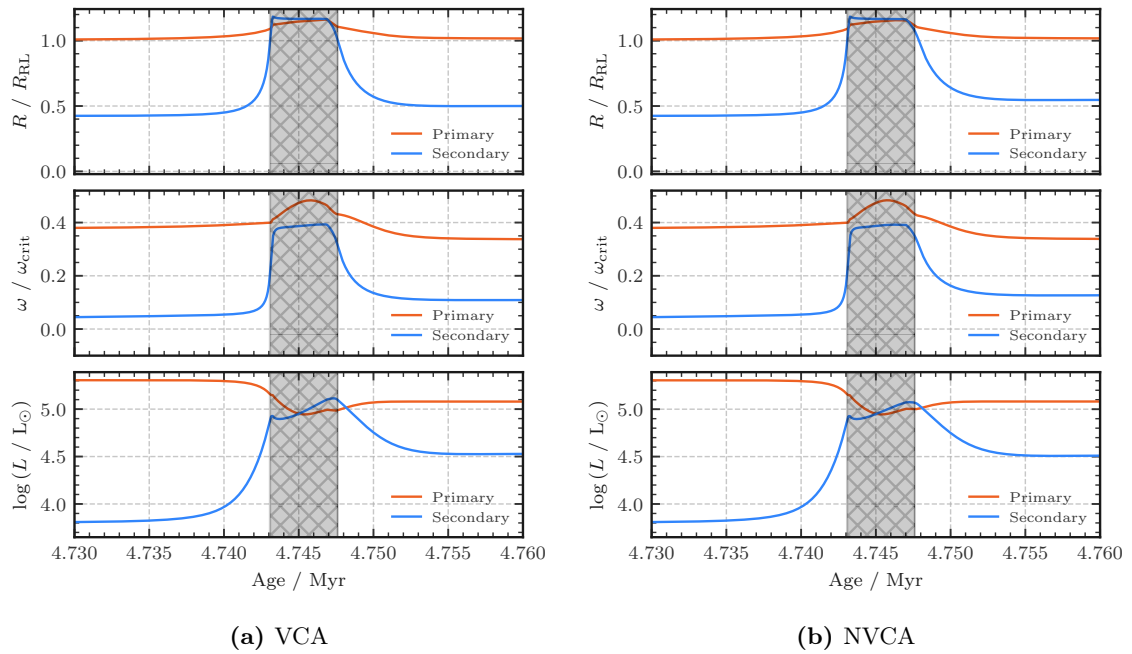


Figure 3.4: The evolution of the radius, rotation rate, and luminosity for the VCA and NVCA models, shown around the contact phase.

3.3.3 | Contact phase (stages C–D)

When the primary star exceeds its Roche lobe during its main sequence, it begins to efficiently transfer material onto the secondary. This mass transfer shrinks the orbital separation of the binary by about 23 per cent, and causes the secondary to swell and fill its own Roche lobe, forming a contact binary. The evolution in the lead up to, during, and immediately after the contact phase is shown in Figure 3.2. Notably, when the system becomes a contact binary, the separation shrinks dramatically owing to conservation of angular momentum. This results in a final period of ~ 2.8 d.

Because of the short initial period of the binary, this stellar model is an example of a system that experiences

a contact phase. This is due to the large amount of mass transferred in the immediate lead-up to this phase, causing the accretor to react by increasing in radius. The process of generating and sustaining the contact phase can be broken down into the following stages, summarised in Figure 3.4:

1. In the immediate lead-up to the contact phase, the accretor begins to accrete material, causing it to spin up. The rotation rate also increases due to transfer of angular momentum and the accretor approaches 40 per cent of its critical rotation rate.
2. This accretion breaks thermal equilibrium in the accretor and causes it to expand (Kippenhahn & Meyer-Hofmeister, 1977; Neo et al., 1977; Lau et al., 2024)
3. The donor dims slightly as the accretor brightens by 63.4 per cent.
4. The donor then begins to increase in luminosity, but its radius returns to approximately the size it was prior to the onset of RLOF, which is below the expected radius for a star of that luminosity.
5. The disruption of thermal equilibrium in the accretor only lasts for some time until the accretor's thermal time-scale increases so it can equilibrium. This means that the contact phase lasts for 5 kyr.

During the contact phase, the accretion of material onto the accretor is less efficient. The mass gain is limited from the the balance between its own mass loss due to filling its Roche lobe and the mass gain from the donor star. The mass transfer efficiency during this phase drops. We can see the instantaneous mass transfer efficiency fluctuates during the contact phase in Figure 3.2. The efficiency drops to below 50 per cent during the contact phase.

Eventually, the accretor shrinks within its Roche lobe and the contact phase terminates. The reason for this is that the accretor's mass has increased to such a point that its thermal timescale is now equivalent to the mass transfer rate from the donor star as shown in Figure 3.5. The accretor is thus able to achieve thermodynamic equilibrium and return to its more compact state. This shrinking is not due to the heavier helium-rich material that has been accreted, although as we note below, post-contact the accretor with VCA is smaller in radius due to this effect, as shown in Figure 3.6 and Table 3.2.

3.3.4 | Post-contact (stages D–E)

There is a clear and consistent difference between the post-contact models. As shown in the inset of Figure 3.6, the models diverge shortly before the termination of the contact phase. At this point, however, the process has become irreversible. The VCA model has become 0.01 dex, or ~ 1.16 kK hotter than the NVCA model. This in turn leads the VCA accretor to be hotter and eventually more luminous than its NVCA counterpart, a theme that is reflected once the post-interaction behaviour is to be studied. The changed temperature/luminosity profile is due to the changed opacity of the star given the helium-rich accretion; since the opacity scales like $\kappa = \kappa_0 (1 + X)$, if the hydrogen mass fraction drops then the opacity increases, increasing the gas pressure (since $P_{\text{gas}} = \frac{2}{3} \frac{g}{\kappa}$), thus driving the temperature and luminosity higher.

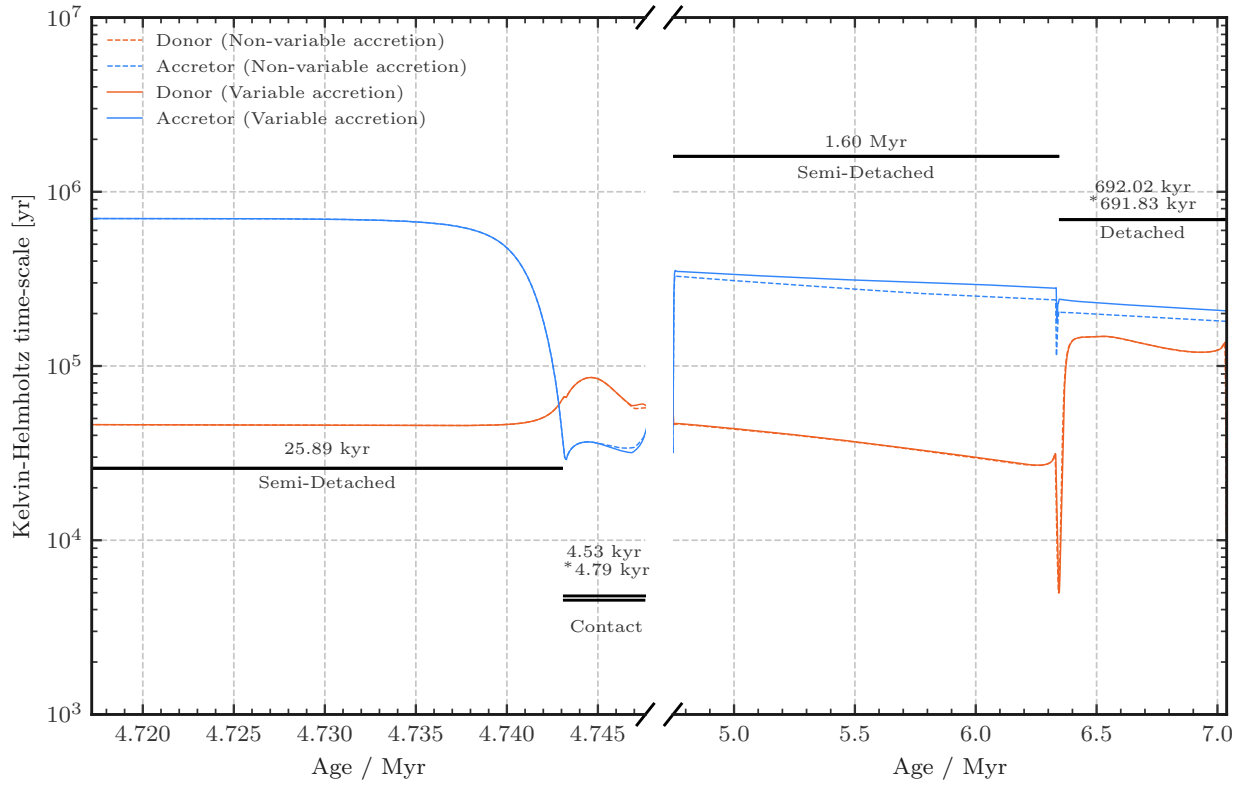


Figure 3.5: The Kelvin-Helmholtz time-scales of both stars in the VCA and NVCA regime. VCA prolongs the final detached phase and shrinks the contact phase. The solid black lines show the length τ of each phase – thermal equilibrium is broken if $\tau_{\text{KH}} > \tau$, i.e., in the first semidetached phase and the contact phase. In the contact and detached phases, the asterisk represents the duration of the NVCA phase.

We thus examine the reasons for this bifurcation. In Figure 3.7 we show the internal composition of the two stars in each binary at each timestep. Noticeably for stage (D), the accretion of helium-rich material has introduced a layer of helium *atop* a layer of hydrogen, in the VCA case. This hydrogen is the original photosphere of the accretor, and the material has not yet mixed in at this stage, due to the continued injection of helium from the donor star.

Eventually, the extended envelope of the donor star cannot be supported, as it has lost a substantial amount of material. This leads to the collapse of the envelope, and the donor becomes a compact helium star. Over the contact phase, the average mass transfer efficiency has been 62.62 per cent, with the donor star losing $7.81 M_{\odot}$ of material and the accretor star gaining $4.89 M_{\odot}$ of material. The balance is assumed to be lost to the ISM. The instantaneous efficiency, shown in Figure 3.2 and Figure 3.3, peaks at 100 per cent immediately prior to the start and end of the contact phase, and post-contact stays that way for ~ 5 kyr after contact terminates as the system settles into a stable state with roughly constant radii, masses, and separation.

3.3.5 | Post-interaction and future history (stage E–onwards)

After the accretor detaches from its Roche lobe, its internal structure is quite different. In both the VCA and NVCA models, the donor has a hydrogen-exhausted core with a hydrogen-poor envelope. However, in the

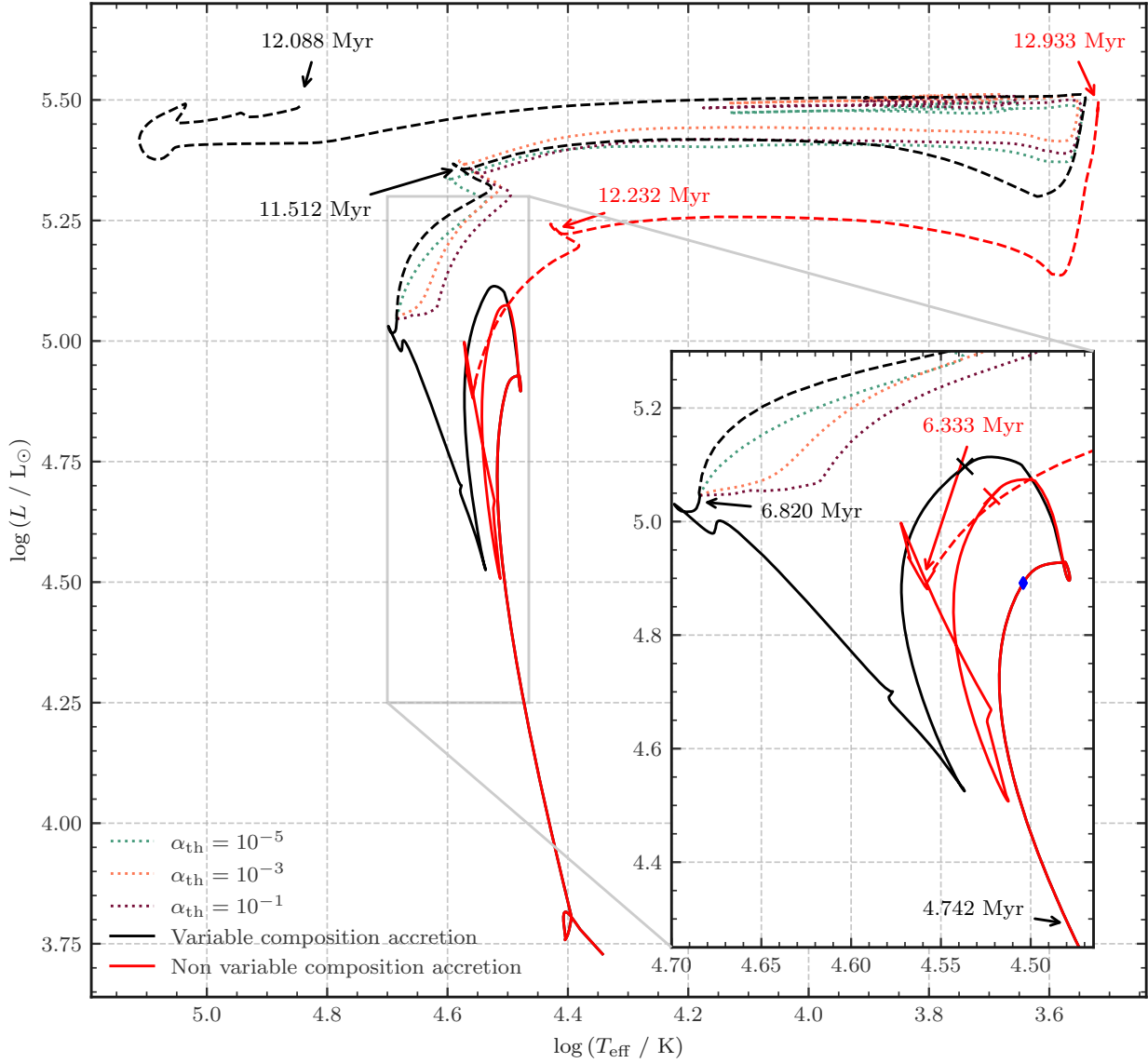


Figure 3.6: An HRD showing the evolution of our model both with (black lines) and without (red lines) the VCA prescription. In the inset, the diamond shows the start of the contact phase (identical for both models), and the coloured crosses show the end of the contact phase for their respective models. The dotted lines show models with both thermohaline mixing and VCA enabled. Models with thermohaline mixing enabled were computed with 199 meshpoints, whereas ones without were computed with 999 meshpoints. The thermohaline mixing models undergo breathing pulses prior to model termination.

VCA model, the profile of the accretor is more complex. As shown in Figure 3.7 (stage D), the accretor has a helium-rich core ($Y_{\text{core}} \sim 0.35$) from the result of ongoing core hydrogen burning, a hydrogen-rich lower-envelope region above the core, with an outer-envelope composition profile of increasing helium abundance towards the surface. The formation of helium stars must require some amount of helium-rich mass transfer to a companion, thus, such accretor structures must be common among the companions of observed helium dwarfs (cf. Section 3.6.3).

The evolution of these composition gradients is determined by thermohaline mixing and rotationally induced mixing, as discussed by [Renzo & Götberg \(2021\)](#). Firstly, thermohaline mixing allows the helium-rich material to mix into the stellar interior, although the time-scale of this process is uncertain. We discuss the impact of this below. Second, if the star is rapidly rotating, rotationally induced mixing (which is not included in these models) may induce further mixing of material to smooth out the composition gradient (e.g. [Eggenberger et al., 2010](#)). As noted in [Renzo & Götberg \(2021\)](#), these mixing processes create a large level of complexity in the structure and evolution of the secondary, with the speed with which the composition gradient can be removed. However, there is still an impact on the evolution of the companion given that the average mean molecular weight of the star increases because of the extra helium that has been transferred.

Finally, we note that all models with thermohaline mixing enabled were evolved with 199 meshpoints, compared to the 999 for the models without thermohaline mixing. The models with thermohaline mixing experience breathing pulses prior to model termination. We tested evolving these models with 199, 499, and 999 meshpoints to determine firstly if the models were sensitive to the number of meshpoints, and secondly to see if more meshpoints would allow us to evolve through the breathing pulses, which are typically considered to be a numerical artefact ([Castellani et al., 1985](#); [Stancliffe & Eldridge, 2009](#), e.g.). We find that the models do not depend on the number of meshpoints, and that we are able to evolve them through breathing pulses using this method. However, since they do not differ significantly, and crash shortly these pulses for unrelated reasons, for the clarity of the plot we omit these from the diagram.

3.4 | Atypical abundance profiles

As previously discussed, the accretor star has an atypical mean molecular weight profile. The discrepancy can be most seen in Figure 3.7. If thermohaline mixing is included, it would at this stage act to attempt to homogenise the star, with the heavier helium material effectively sinking to the core of the star. The effect of this will vary depending on the efficiency of thermohaline mixing, which is not conclusively known. To that end, the model of [Eggleton \(1972\)](#); [Kippenhahn et al. \(1980\)](#); [Stancliffe et al. \(2007\)](#) is employed, where the free parameter α_{th} controlling the strength of thermohaline mixing is allowed to vary. $\alpha_{\text{th}} = 0$ corresponds to no thermohaline mixing, whereas $\alpha_{\text{th}} = 1$ represents strong (virtually instantaneous) thermohaline mixing.

We show in Figure 3.9 the effect that thermohaline mixing has on the accretor. Varying the thermohaline mixing coefficients allows for the shape of the HRD track to change noticeably, with the locations of the hook changing, as well as the trajectory as the model crosses the Hertzsprung gap. However, and critically, even

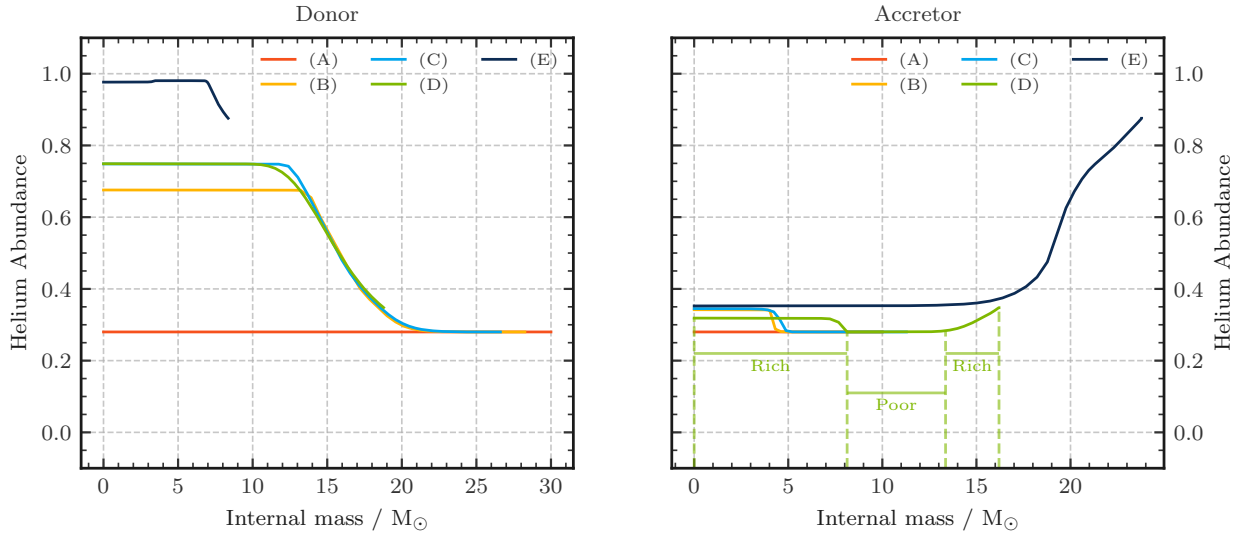


Figure 3.7: The abundance of helium at the five critical stages of the binary evolution. The green lines under the accretor's plot show the regions where the accretor is helium-rich and helium-poor.

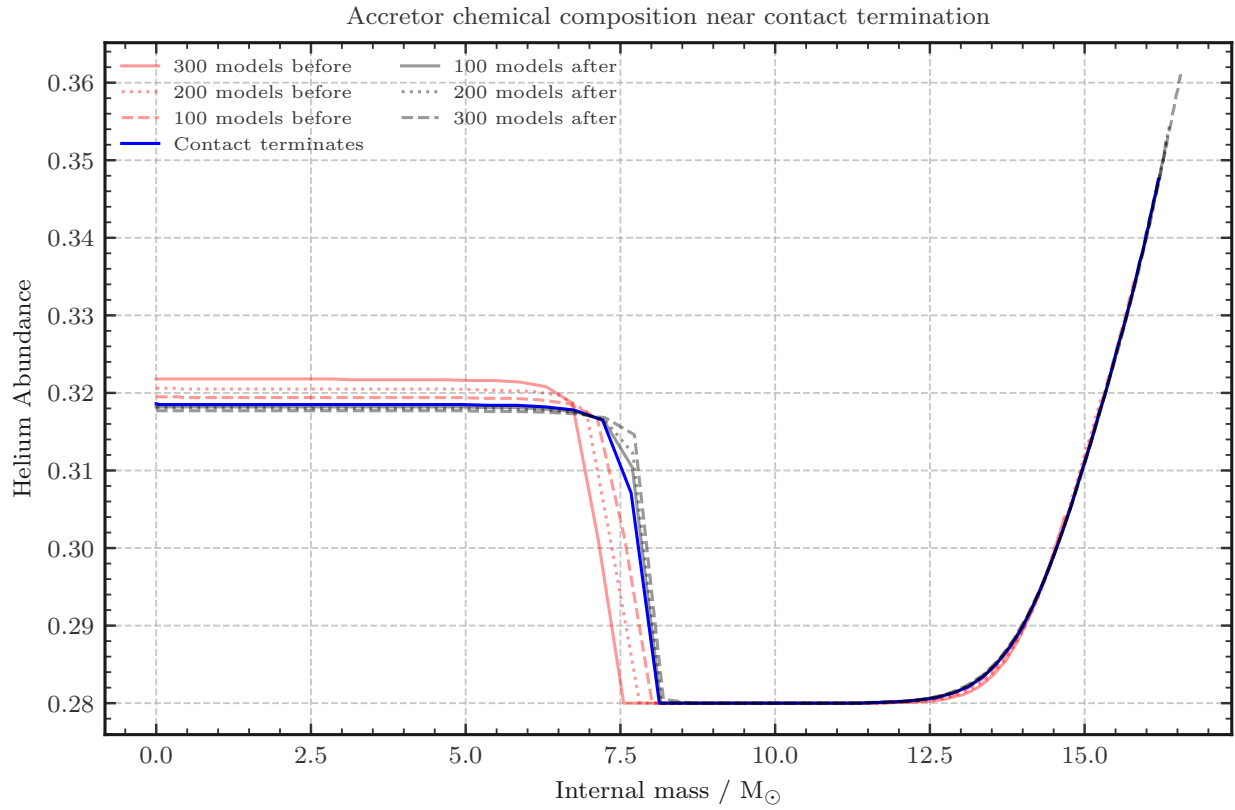


Figure 3.8: The helium abundance in the accretor near when the model detaches. The star contracts as mass transfer ends, resulting in a $\sim 9 R_{\odot}$ accretor 300 models after collapse, compared to $\sim 11.5 R_{\odot}$ 300 models before (a difference of ~ 1.6 kyr).

the most efficient thermohaline mixing included – $\alpha_{\text{th}} = 0.1$ – was unable to return the model to its location on the HRD, even though it homogenised the model only within a few timesteps.

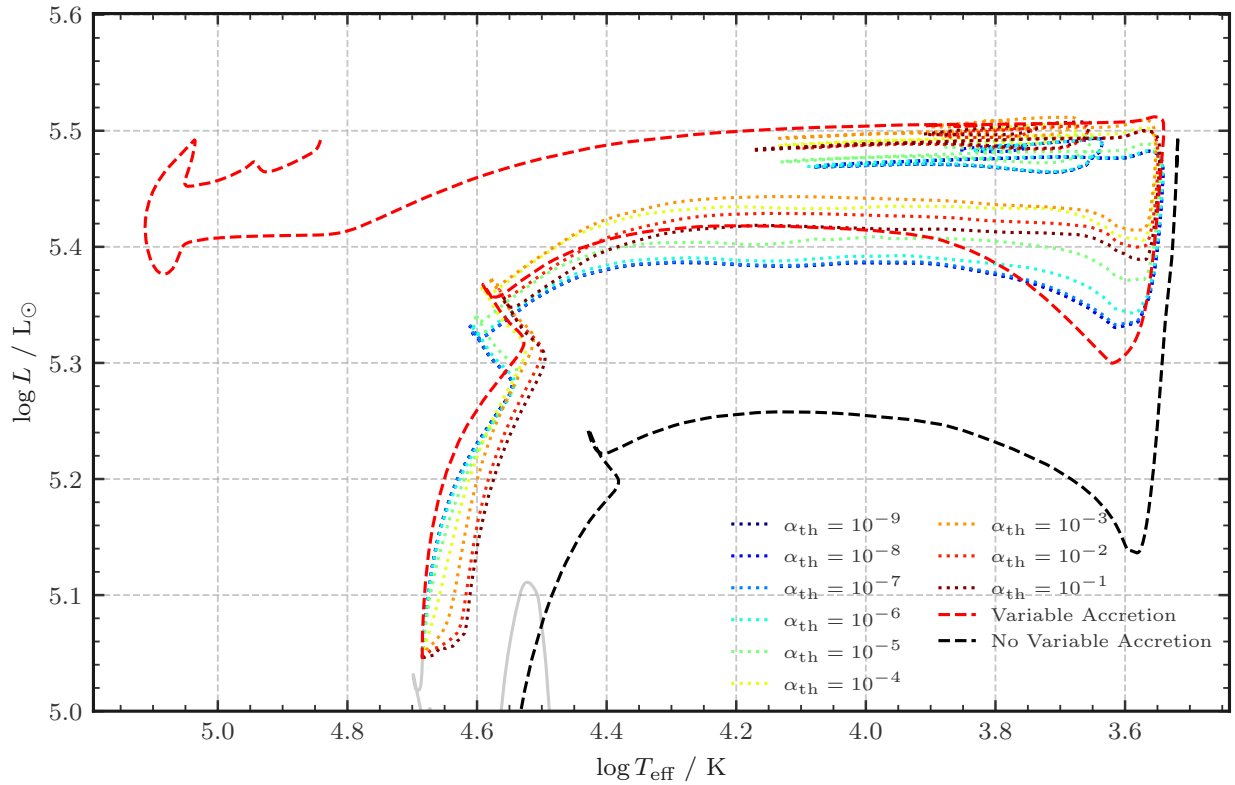


Figure 3.9: An HRD of the post-mass transfer secondary model evolved as a single star with different thermohaline mixing coefficients. The diagram has been truncated to have the y -axis begin at $\log L = 5$, with the input model shown in faint grey. The “variable accretion” and “non-variable accretion” models were evolved with thermohaline mixing entirely disabled and with 199 meshpoints. The thermohaline mixing models were evolved with 999 meshpoints, and all experience breathing pulses prior to model termination.

3.5 | Comparison between VCA and NVCA

In Figure 3.6 we display a comparative HRD between the cases of VCA and NVCA. In each case an $30+10 M_{\odot}$ binary was evolved with the only difference between the two being whether material was allowed to have VCA or NVCA.

The first and most striking conclusion that we can draw from this result is that the companion is, in the VCA case, hotter and more luminous than the NVCA case. As previously mentioned, even near-instantaneous thermohaline mixing to homogenise the star was insufficient to return the accretor to a position on the HRD typified by NVCA models.

Indeed, none of our single-star models are able to replicate the NVCA model. However, a $30 M_{\odot}$ model has a similar luminosity to the accretor but its surface temperature is 0.15 dex ($\sim 9.6 \text{ kK}$) higher. This is due to the lower opacity of the helium-rich surface of the accretor. Similarly, we see that during the main sequence the accretor with VCA is also approximately 0.15 dex ($\sim 11.3 \text{ kK}$) hotter in surface temperature than the NVCA model.

		$t = 0$ (A)	Star 1 RLOF		Star 2 RLOF		Star 1 RLOF		Star 2 RLOF	
			Begin	End	Begin	End	Begin	End	Begin	End
			(B)	(E)	(C)	(D)	(B)	(E)	(C)	(D)
Donor	$M / M_{\odot}$	30.00	27.93	7.56	26.91	17.43	27.93	7.57	26.91	17.27
	$R / R_{\odot}$	8.82	16.32	10.20	13.88	9.76	16.32	10.15	13.88	9.77
	$\log L / L_{\odot}$	5.07	5.30	5.25	5.16	4.94	5.30	5.25	5.16	4.98
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.09	0.70	0.57
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.89	0.28	0.42
Accretor	$M / M_{\odot}$	10.00	9.99	20.97	11.00	14.57	9.99	20.77	11.00	14.52
	$R / R_{\odot}$	4.87	4.36	4.02	9.23	8.98	4.36	6.53	9.23	9.02
	$\log L / L_{\odot}$	3.73	3.81	4.89	4.90	5.06	3.81	4.73	4.90	4.95
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.70	0.70	0.70
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.28	0.28	0.28
Binary	$P_{\text{bin}} / \text{d}$	3.98	3.85	4.41	3.14	2.52	3.85	4.38	3.14	2.54
	$a / R_{\odot}$	36.14	34.74	34.56	30.28	24.71	34.74	34.33	30.28	24.78
	Age / kyr	0.00	4707.16	6450.23	4712.75	4716.42	4707.16	6448.11	4712.75	4716.70

(a) VCA
(b) NVCA

Table 3.2: The key evolutionary parameters of both stars and the binary configuration at five critical points: the initial parameters, and the beginning and ends of both epochs of RLOF. All values are rounded to 2 decimal places. The letter in parentheses represents which stage of Figure 3.7 the abundances are given in.

Within the VCA model, the formation of an inverted mean molecular weight gradient means that thermohaline mixing must be considered. In Figure 3.6 we can see that higher values of α_{th} lead to a more rapid decrease in surface temperature by approximately 0.05 to 0.08 dex. The most efficient thermohaline mixing that we include removes the composition gradient in a few time-steps (see Figure 3.10), but it remains hotter and more luminous throughout the main sequence compared to the NVCA model. This is because while the composition gradient is removed the star has a higher helium abundance. Figure 3.6 shows that when thermohaline mixing is included we see this temperature difference decrease to approximately 0.1 dex (~ 1.54 kK). Thermohaline mixing has a bigger difference on the accretor tracks earlier in the main sequence evolution than at the end of the main sequence, where the tracks are more similar in their evolutionary path.

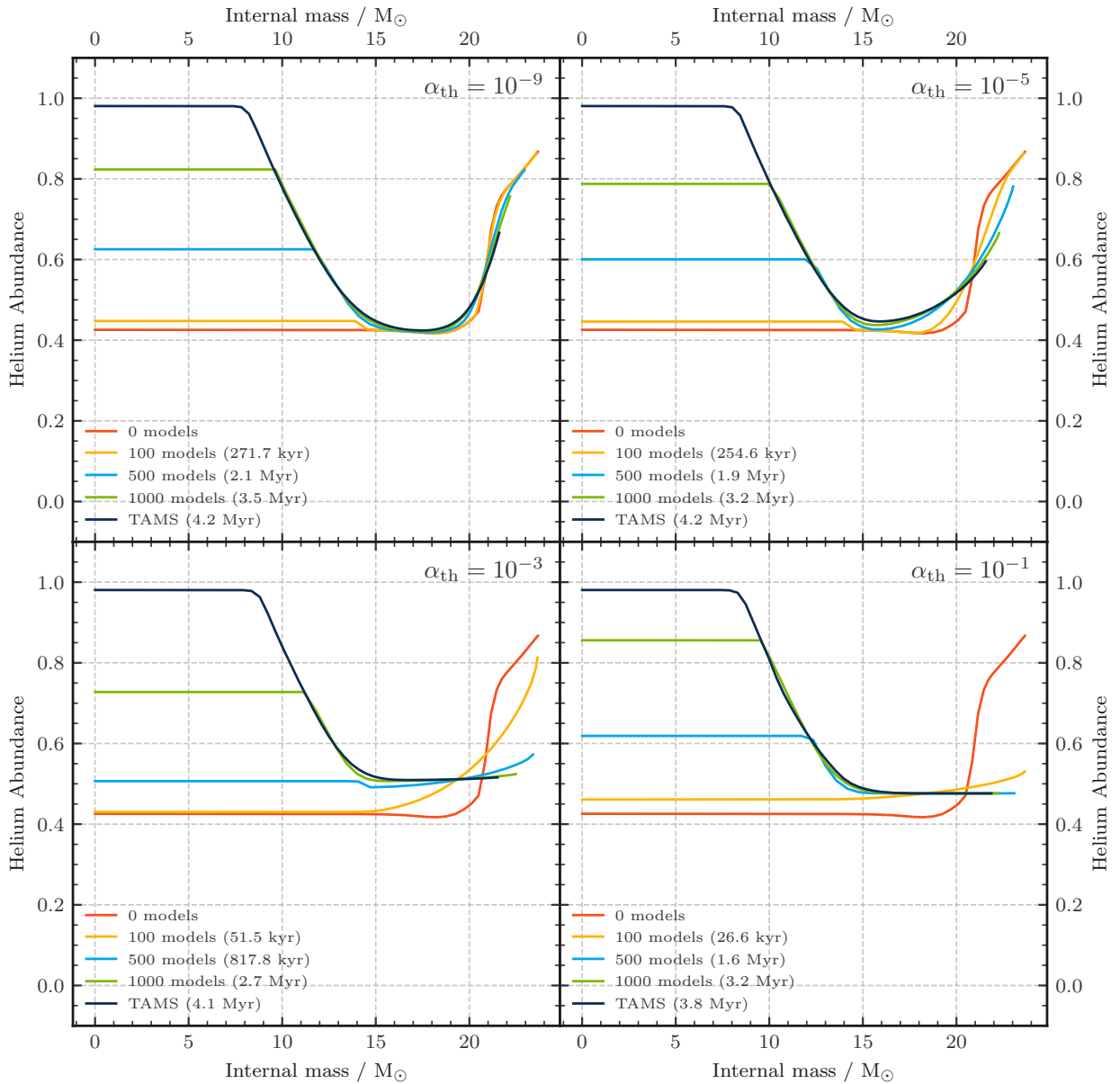


Figure 3.10: The helium abundance profile for each model of thermohaline mixing, as a function of the internal mass of the star. The legend key shows the model number and age corresponding to each line, measured from the beginning of the single-star evolution procedure described in Section 3.3.5. Only the VCA models are shown in this plot. The NVCA models have a flat composition profile at the surface.

3.6 | Implications

An important question to consider is what the VCA model would be observed as. Some of the parameters of the star are typical of single main sequence stars but with higher luminosities, helium-rich abundances, and higher surface temperatures they may be able to be identified from observations. The effective temperature of our VCA models is in the range of O5 to O3 stellar types. The hottest examples might therefore look similar to the WN/O3 stars identified by [Massey et al. \(2023\)](#), although it has previously been suggested that such stars were in fact the helium donors from binary interaction ([Götberg et al., 2018](#); [Smith et al., 2018](#)). The idea that the WN/O3 stars could in fact be the accretor stars does link well with their apparent single-star nature ([Massey et al., 2023](#)) and their relative isolation compared to other massive stars ([Smith et al., 2018](#)), if their donor stars have already exploded in a core-collapse supernova. Alternatively, it is also possible that the accretors may be observed as luminous-blue variable (LBV) stars. [Smith et al. \(2012\)](#) suggested that LBV stars may arise from accretors in binaries.

Given that the evolution of the accretor star remains different from that of a single star, the impact of VCA may provide at least part of the answer to some outstanding questions in stellar evolution. The first question is the known mass discrepancy for some stars between their spectroscopic and evolutionary masses (e.g. [Herrero et al., 1992](#); [Massey et al., 2009](#); [Bouret et al., 2013](#); [Massey et al., 2013](#); [McEvoy et al., 2015](#); [Markova et al., 2018](#); [Ramachandran et al., 2018](#); [Bestenlehner et al., 2020](#)). Comparing the luminosities of our accretor models with single-star models in Figure 3.11, we can see that a star with helium-rich accretion has a luminosity similar to a star that is 20 per cent more massive. Such a difference may explain why for some stars' mass estimates can differ significantly.

The second implication of our VCA model is that the higher effective temperatures of the models lead to more of a star's luminosity being radiated at wavelengths below $912 \text{ \AA}$ which can help drive ionisation of the surrounding media. This has important implications for understanding the nebula emission from the HII regions and reionization (e.g., [Stanway et al., 2016](#); [Eldridge & Stanway, 2022](#); [Ghodla et al., 2023](#)).

Third, work has shown that mass transfer can lead to accretors with less bound envelopes ([Laplace et al., 2021](#); [Renzo et al., 2023](#)), resulting in systems for which the energy required to explode the star in a supernova is decreased compared to an equivalent single-star model. As such, as in this case (where the mass transfer is sufficiently efficient that the final core collapse could give rise to unusual supernovae, i.e., an event that does not fit into one of the common supernovae types, IIP, IIL, IIB, Ib or Ic. The recent comprehensive work by [Schneider et al. \(2025\)](#) shows how the accretors can have very different supernovae compared to single stars. We note that it is accepted that stellar mergers also give rise to unusual supernovae (e.g. [Menon et al., 2019](#)) and future study is required to identify signatures in supernovae to differentiate between an accretor progenitor and the result of a stellar merger.

3.6.1 | Spectroscopic masses

Animals can talk.

— THIRD OF THE SIX IMPOSSIBLE THINGS

Given that the evolution of the accretor star remains different from that of a single star, the impact of VCA may provide at least part of the answer to some outstanding questions in stellar evolution. The first question is the known mass discrepancy for some stars between the mass predicted by models (the “evolutionary mass”) and the mass inferred from spectroscopic studies of stars (the “spectroscopic mass”). Summaries are found in works such as [Herrero et al. \(1992\)](#); [Massey et al. \(2009\)](#); [Bouret et al. \(2013\)](#); [Massey et al. \(2013\)](#); [McEvoy et al. \(2015\)](#); [Markova et al. \(2018\)](#); [Ramachandran et al. \(2018\)](#); [Bestenlehner et al. \(2020\)](#). The reason has been left as either deficiencies in the physics of model atmospheres ([Herrero et al., 1992](#)) or non-included or not well-understood physics ([Bestenlehner et al., 2020](#)).

This phenomenon is not unique to one code or methodology. To name but four, it has been observed in:

1. BONNSAI (work in [Bestenlehner et al. 2020](#), BONNSAI in [Schneider et al. 2014](#)),
2. the work of [Maeder \(1990\)](#) ([Herrero et al., 1992](#)),
3. the Geneva tracks (work by [Markova et al. 2018](#), Geneva code by [Ekström et al. 2012](#)), and
4. the PoWR code (work by [Ramachandran et al. 2018](#), PoWR code in [Gräfener et al. 2002](#); [Hamann & Gräfener 2003](#); [Sander et al. 2015](#)).

If luminosities are systematically incorrect, as discussed above, then there would be a corresponding systematic incorrect calculation of the mass; that is, a traditional mass-luminosity relationship may not hold. We show in Figure 3.11 the relationship between the evolutionary mass and luminosity for both VCA and NVCA cases, as evolved in subsection 3.3.5. We also display this for integer mass single-star models. Unsurprisingly, the NVCA case evolves like a $24 M_{\odot}$ model – the initial mass of the model when this single-star evolution began was $23.71 M_{\odot}$, and the abundance profile typifies a single star; since the mass transfer was hydrogen-dominated, the profile is not inverted and thus resembles a single star.

On the other hand, the VCA case would (if observed) initially appear to be more massive than a $24 M_{\odot}$ star (despite being $23.66 M_{\odot}$ initially), eventually appearing to be approximately $21 M_{\odot}$ – i.e., $\sim 3 M_{\odot}$ less massive than it actually would be. Thermohaline mixing suppresses this change somewhat, but even with stronger thermohaline mixing still ultimately results in an accretor that is systematically different to a NVCA model.

Consequently, any masses inferred from spectroscopy or luminosity measurements may be an underestimate of the true (evolutionary) mass. This would affect, for example, the inferred masses of compact objects and thus may have implications for our understanding of the progenitors of gravitational waves – were we

to systematically underestimate the masses of compact objects, then calculations of the compact object coalescence rates (cf. 4) would be incorrect.

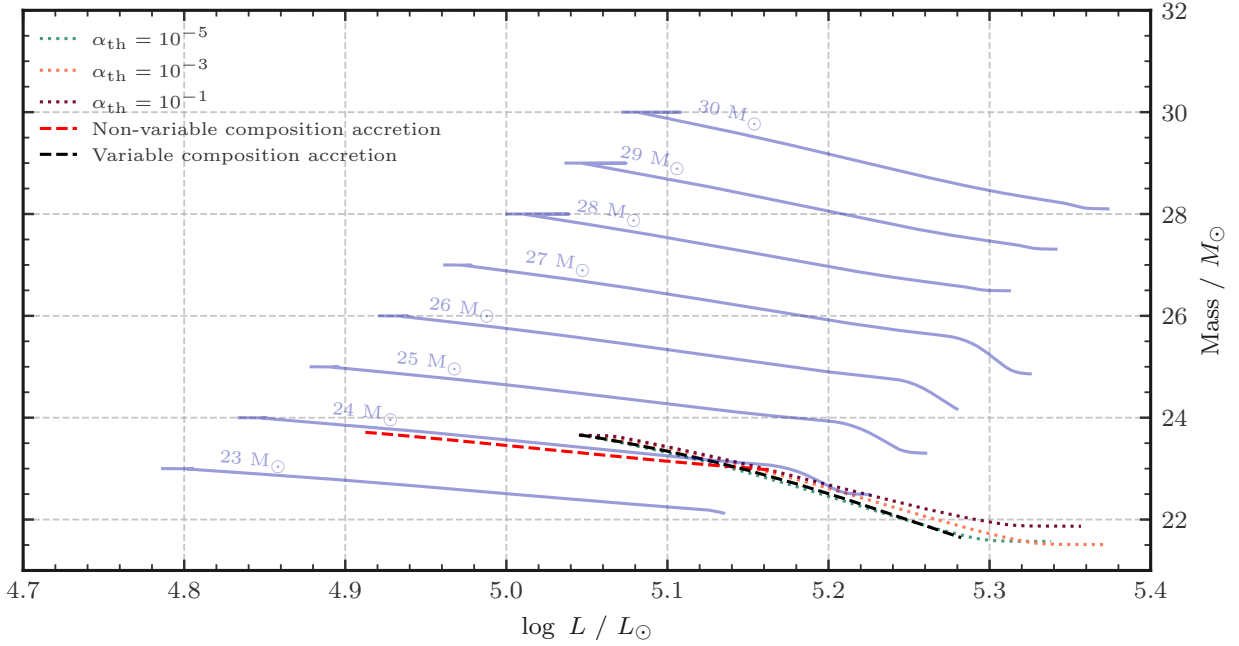


Figure 3.11: The (evolutionary) mass-luminosity relationship in the presence and absence of VCA. Also shown are the curves for reference models of between $M_{\text{ZAMS}} = 21 M_{\odot}$ and $26 M_{\odot}$, which were evolved as a single star.

3.6.2 | Supernovae

Previous work has shown that stellar rejuvenation leads to accretors with less bound envelopes (Renzo et al., 2023), resulting in systems for which the energy required to eject the envelope in a supernova is decreased compared to an equivalent single-star model. As such, rejuvenation such as in this case (where the mass transfer is sufficiently efficient that the accretor becomes rejuvenated) could give rise to unusual SNe. SN 1987A, the most widely studied supernova, is often conjectured to be the result of a binary merger (e.g., Podsiadlowski, 2017) due to its unusual explosion as a blue supergiant instead of a red supergiant, as predicted by theory. A rejuvenated star would appear younger, providing a potential avenue of formation for unusual stellar transients.

The surface temperature of the accretor has important implications for the observations of SN progenitors. The donor in our presented model is massive enough to explode as a Type Ib/c SN. A number of observations exist of the pre-SN site of many Type Ib/c SNe (Eldridge et al., 2013; Eldridge & Maund, 2016). One issue is that in many of the observations not only is the progenitor unseen, so too is a massive main sequence star companion. We note that the higher surface temperature of the accretor makes it more difficult to detect in the optical bands which normally exist for this work. The different structure also makes it more difficult to predict the impact of the SN on the companion (Hirai et al., 2018). Thus another reason to study the accretors in more detail is to understand the progenitor detections of Type Ib/c SNe when so few companions

have been detected in post-SN observations (Eldridge & Maund, 2016).

3.6.3 | The prevalence of helium-rich mass transfer

Although we have presented only one example of helium-rich mass transfer in this work, it is not unreasonable to assume that this process may be common. To estimate how often such evolution may occur, we have used the binary star population of models from the Binary Population and Spectral Synthesis (BPASS) v2.2 code suite (Eldridge et al., 2017; Stanway & Eldridge, 2018). BPASS only models one of the stars in a binary in detail at a time, but enable us to understand how many stars may experience helium-rich mass transfer.

We used an initial metallicity of $Z = 0.020$ for the models, including every binary where the initial mass of the donor is greater than $7 M_{\odot}$ and the accretor accretes more than 0.1 per cent of its initial mass. For an instantaneous burst of a $10^6 M_{\odot}$ with an upper mass limit of $300 M_{\odot}$, an initial mass function from Kroupa (2001) and initial binary parameter distribution from Moe & Stefano (2017), we find that 3543 ± 68 binary stars undergo this phase of evolution, of which 34 ± 1 per cent accrete more than 5 per cent of their initial mass. Of these 3543 ± 68 models, 23 ± 1 per cent accrete material, which reaches a maximum helium abundance of greater than 0.8. Thus, the evolution of such accretors is common and is of vital importance to understand.

3.7 | Chapter Summary

We have exhibited a case study of a close $30 + 10 M_{\odot}$ binary, in which mass transfer becomes dominated by helium. By contrasting this to a binary with hydrogen-dominated mass transfer, we have shown that this results in an atypical composition profile that produces unexpected accretors. These accretors have systematically different luminosity compared to their hydrogen-dominated alternatives, translating to a different (incorrect) inferred mass.

In summary, the key conclusions of this chapter are as follows:

1. If a close binary system has a helium star within it, it can only have been formed by binary interactions. The accretor is likely to have accreted some helium-rich material, and as shown by our VCA model, the accretor is hotter and more luminous than if the transferred material was not helium-rich. In addition, the eventual fate supernova time of the model is altered from a hydrogen-rich type II supernova to a hydrogen-free type Ib/c supernova.
2. Accretor stars with a helium-rich surface layer on top of a hydrogen-rich lower envelope and helium-rich convective core can be very different to a single star of the same mass. In the case presented, a non-thermal-equilibrium $24 M_{\odot}$ accretor star was 0.2 dex more luminous and 12.5 kK hotter than a single star of the same mass. This means that in a binary system post-mass transfer, normal mass-luminosity relationships may not hold for accretor stars, and stellar masses may be overestimated from luminosity.
3. When thermohaline mixing is included in these models, the surface composition gradient is reduced. This leads to the accretor becoming cooler by 0.1 dex but it remains overluminous for its mass having

the luminosity of a $30 M_{\odot}$ single star despite being only $24 M_{\odot}$. The strength of the thermohaline mixing determines how quickly the accretor stars achieve a lower surface temperature and impacts the lifetime on the main sequence.

4. The mass transfer efficiency in this binary interaction evolves with time through the mass transfer phase. A contact phase of evolution limits the mass transfer efficiency because the secondary cannot accrete more material whilst filling its Roche Lobe. But later, once the contact phase ends, the accretor becomes massive enough for its thermal time-scale to come close to the mass transfer time-scale, mass transfer efficiency increases.
5. We used BPASS v2.2 to perform a population synthesis and calculate the expected number of systems that experience helium-rich mass transfer. At Solar metallicity, 23 per cent of all systems in which mass transfer occurs, accrete helium-rich material with a maximum helium mass fraction of greater than 0.8. Suggesting helium-rich mass transfer is a common and important process to account for in models of binary evolution.

VCA has a clear influence on the future evolution of an accretor in a binary star system. What remains to be seen is the influence on a population writ large; the effect of VCA should be included in binary population synthesis projects to understand the broader implications of VCA. We have suggested that these could be to explain some of the discrepancy between spectroscopic and evolutionary masses of massive stars, lead to greater ionising output for a stellar population and possibly explain some exotics stellar objects such as LBVs. However, the biggest uncertainty is how efficient thermohaline mixing is in such stars. However, observations such as those demonstrated by [Miszuda et al. \(2021\)](#) of stellar oscillations in post-interaction binary stars will be key to understanding this physical process.